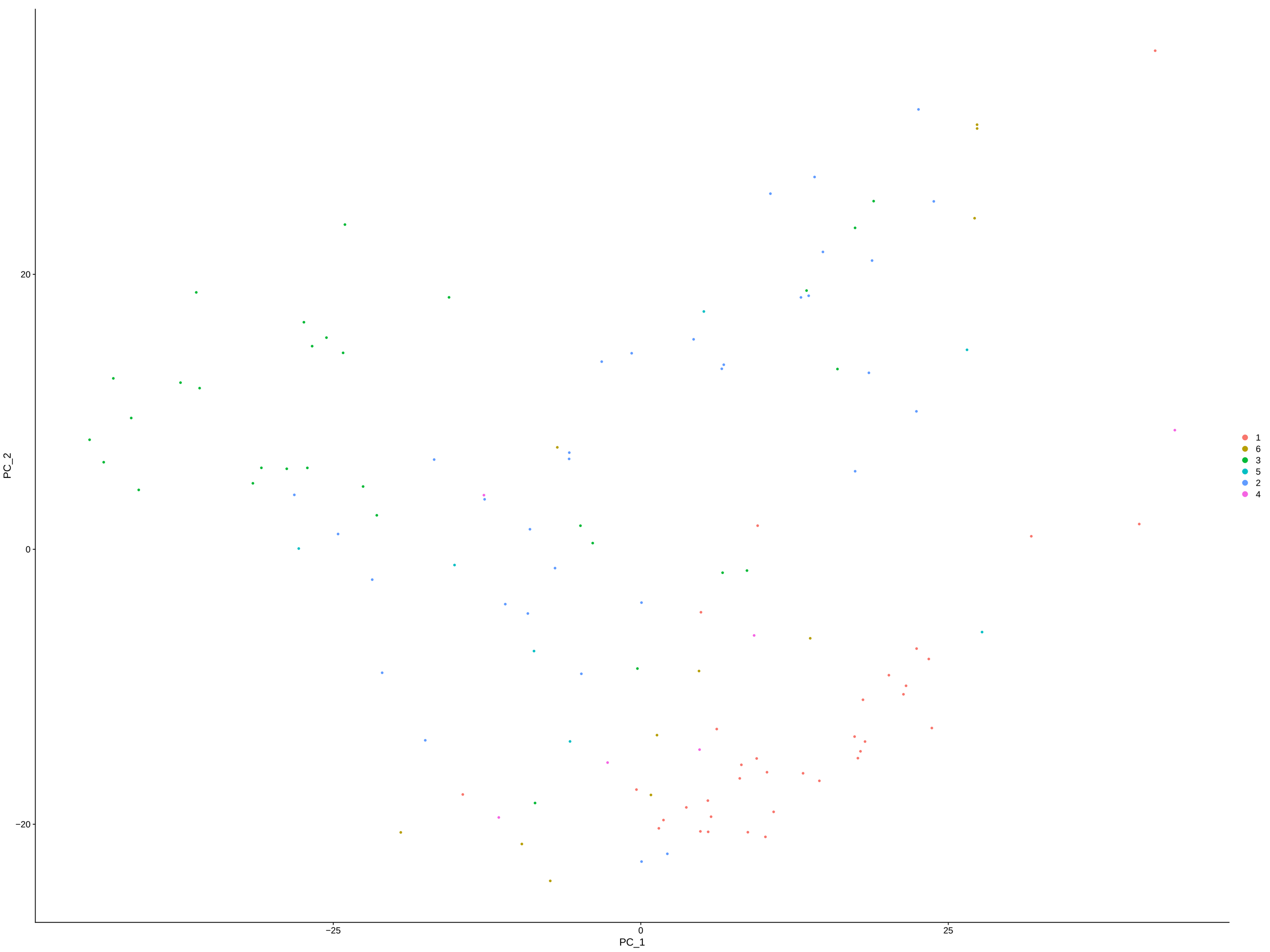
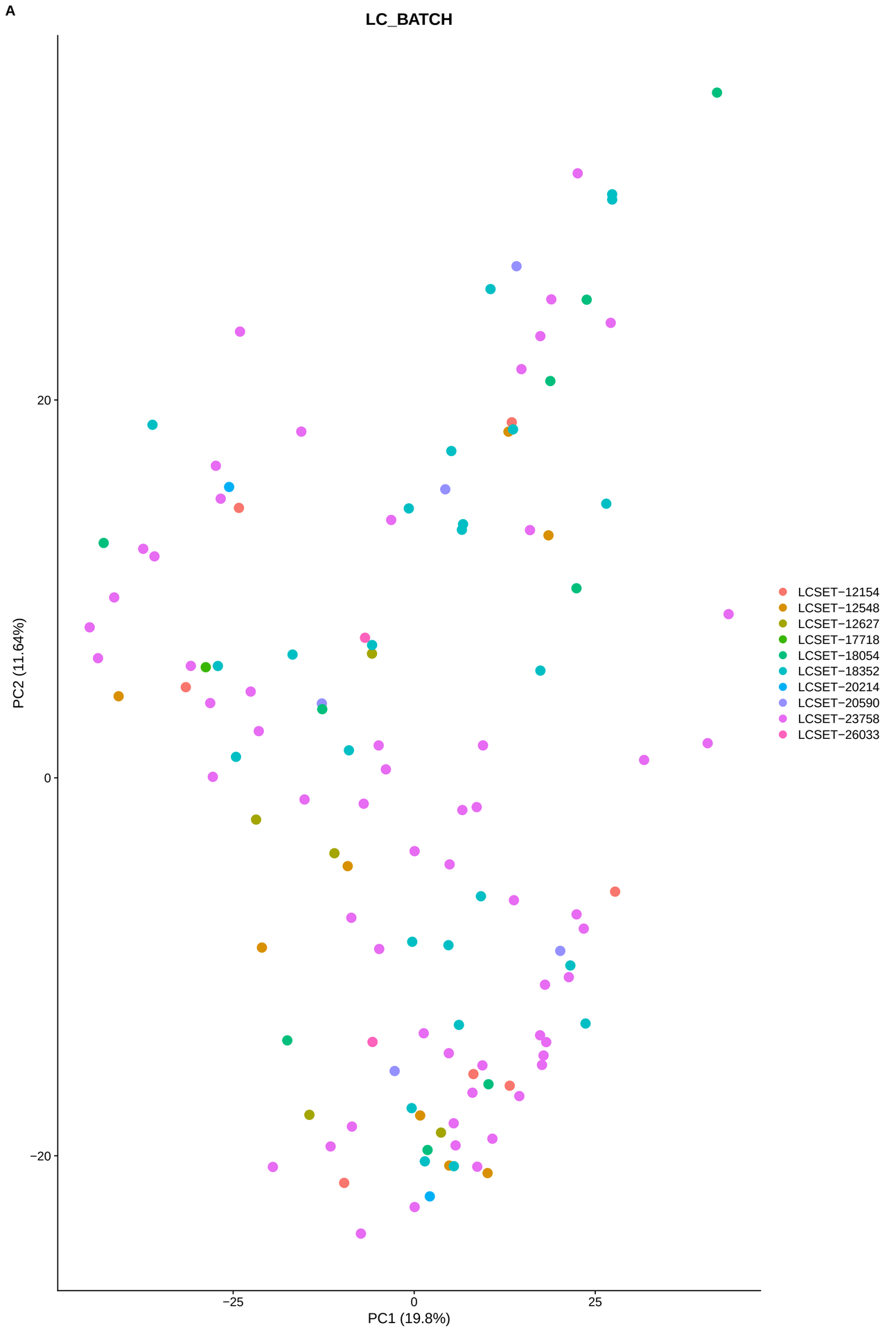
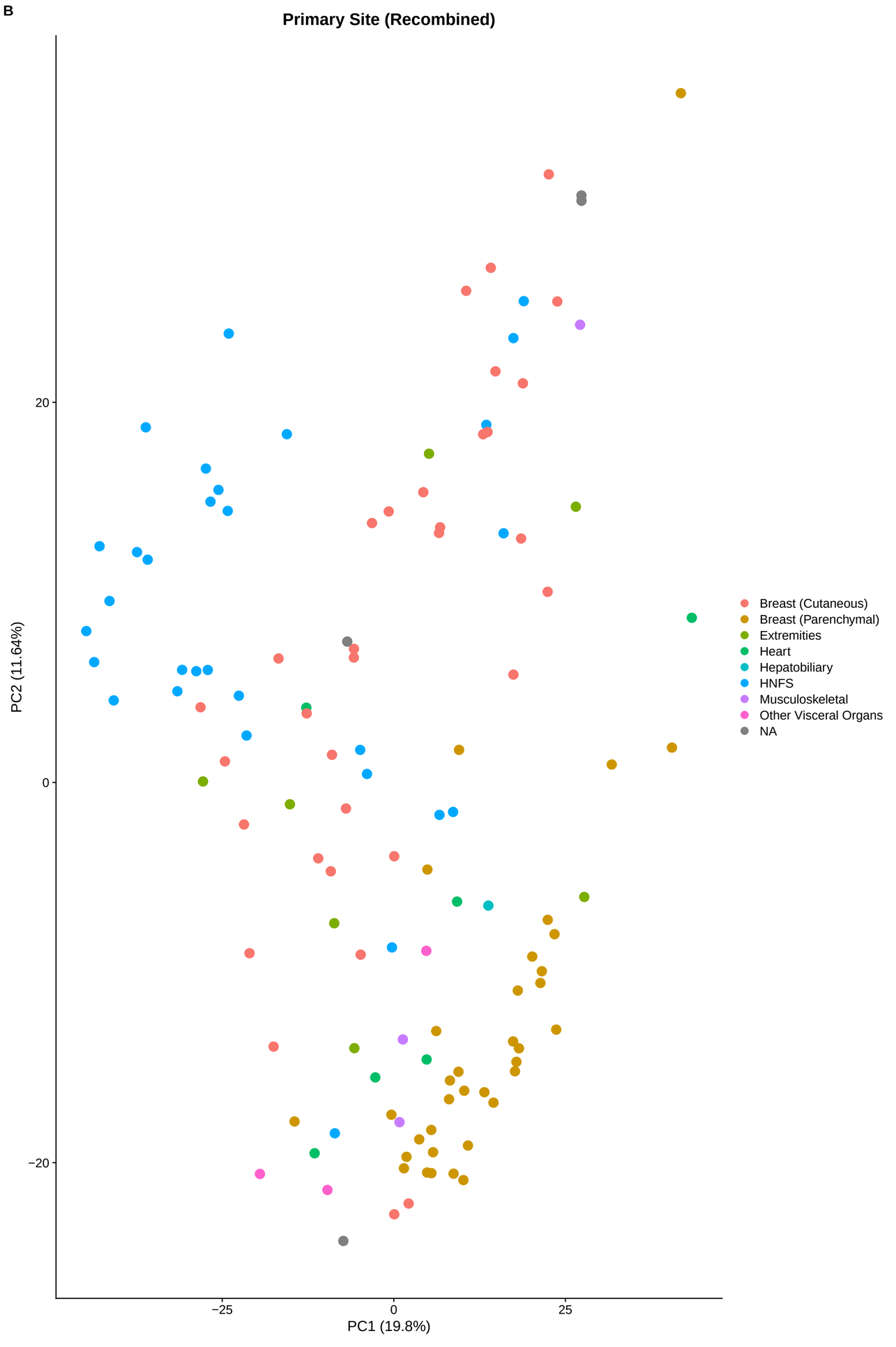
Cluster 1
Cluster 2
Cluster 3
Cluster 4
Cluster 5
Cluster 6
Primary Site
Cutaneous
RAAS/LAAS
Age
Sex
Epithelioid
Spindle Cell
Nuclear Grade
Angiogenesis
NID2
RCBTB2
UBTFL6
FLT1
PKD1L1
CYYR1
PABPC4L
KCNC4
CPNE5
COL15A1
Lymphangiogenesis
MYC
MMP17
TRAP1
PRSS21
POTEE
ELAC2
TSR1
MAGEB6P1
RPL22L1
ARHGAP42P1
FGFR Enriched
ERVMER61−1
CASK
KRT34
KRT85
PIP4K2C
HTN1
KRTAP24−1
ELOVL3
MSMO1
PTPN3
17q Amplified
TCF21
GATA6
HHIP
C4B
CCDC68
C7
SGMS2
VEGFA
C4A
ANKRD42
Radioresistant
HOXD11
CDH20
PTPRN
DUSP9
ZIC5
HOXD10
TIMM8AP1
DISP2
UCHL1
HILS1
Repr. Somatic Mut.
Repr. Germline Var.
TMB
Gene log2FC
Gene Significance
Gene log2FC
6
4
2
0
Gene Significance
adj. p < 0.01
nom. p < 0.05
Primary Site
Breast (Cutaneous)
Breast (Parenchymal)
Extremities
Heart
Hepatobiliary
HNFS
Musculoskeletal
Other Visceral Organs
Cutaneous
Cutaneous AS
Non−cutaneous AS
RAAS/LAAS
LAAS
Non−RAAS/LAAS
RAAS
RAAS & LAAS
Age
80
60
40
20
Sex
Female
Male
Epithelioid
NO
YES
Spindle Cell
NO
YES
Nuclear Grade
HIGH
LOW
Scaled Expression
3
0
−3
Repr. Somatic Mut.
CFTR
KDR
Not available
Others
PLCG1
POT1
TP53
Repr. Germline Var.
BRCA2
CFTR
CHEK2
No PV Detected
Others
POT1
TP53
TMB
<= 1 mut/MB
<= 10 mut/MB
<= 3 mut/MB
> 10 mut/MB
NA
Gene Set
Angiogenesis
Lymphangiogenesis
FGFR Enriched
17q Amplified
Radioresistant

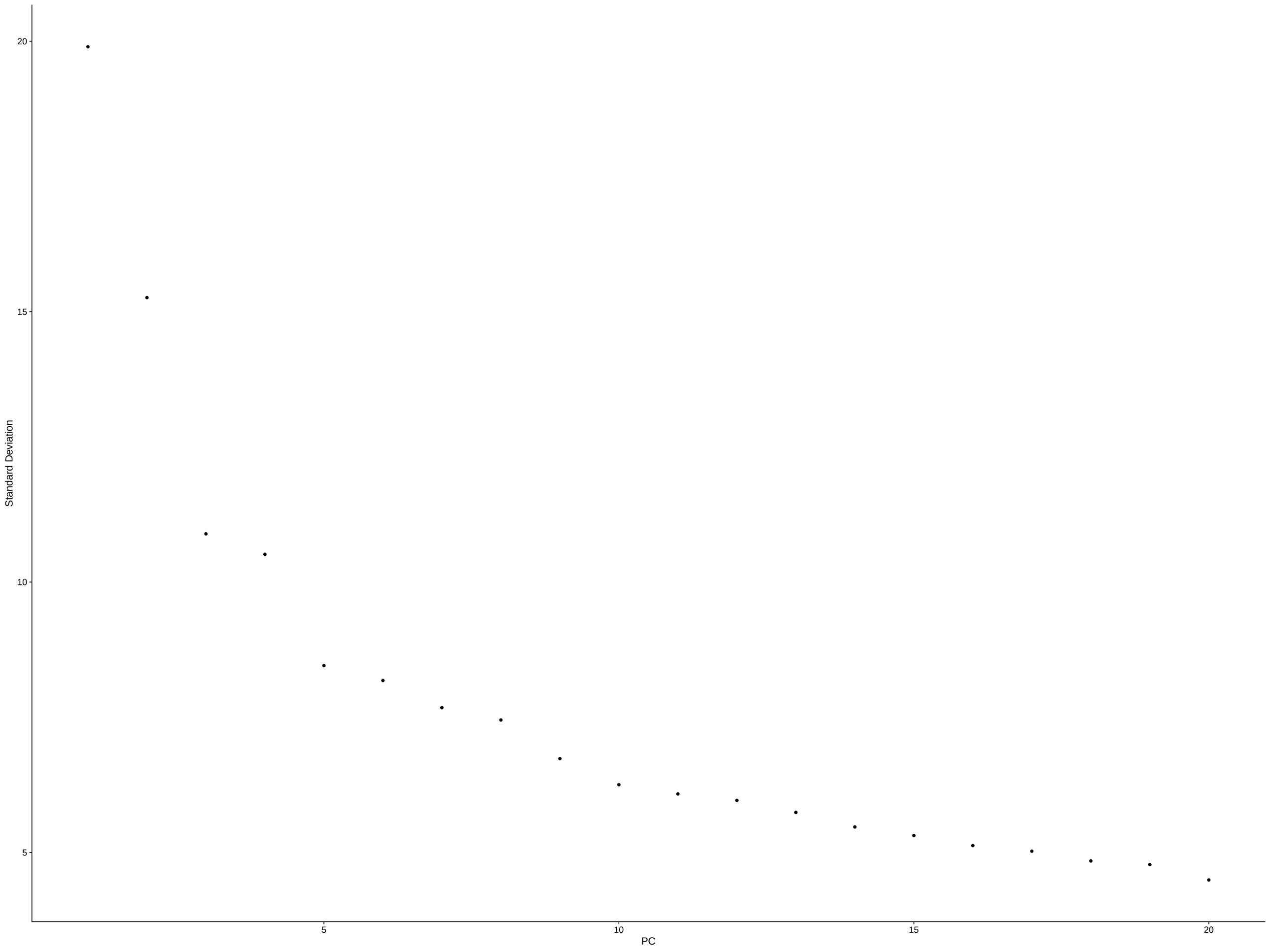

20
0
−20
PC_2
−25
0
25
PC_1
1
6
3
5
2
4

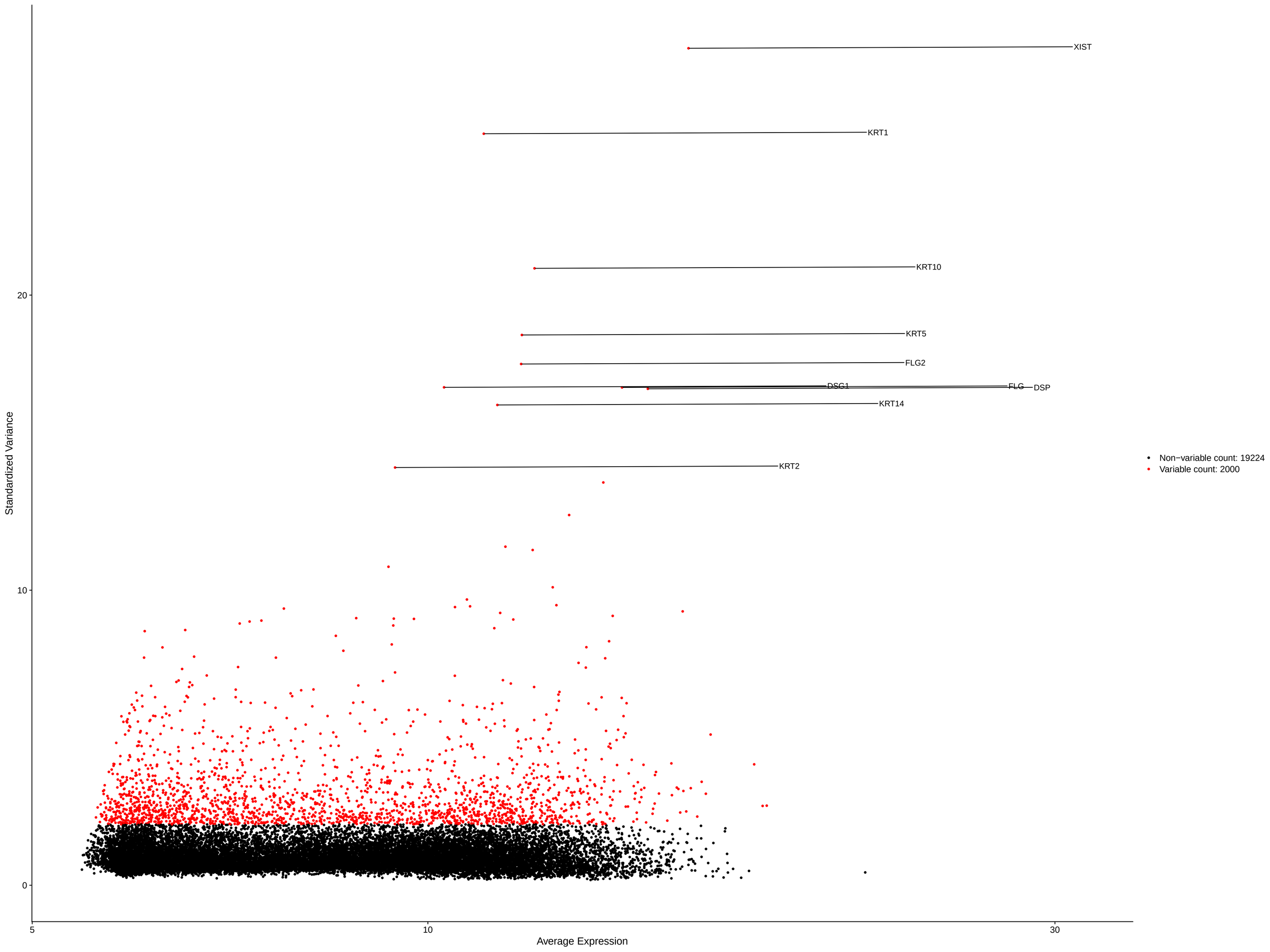
A

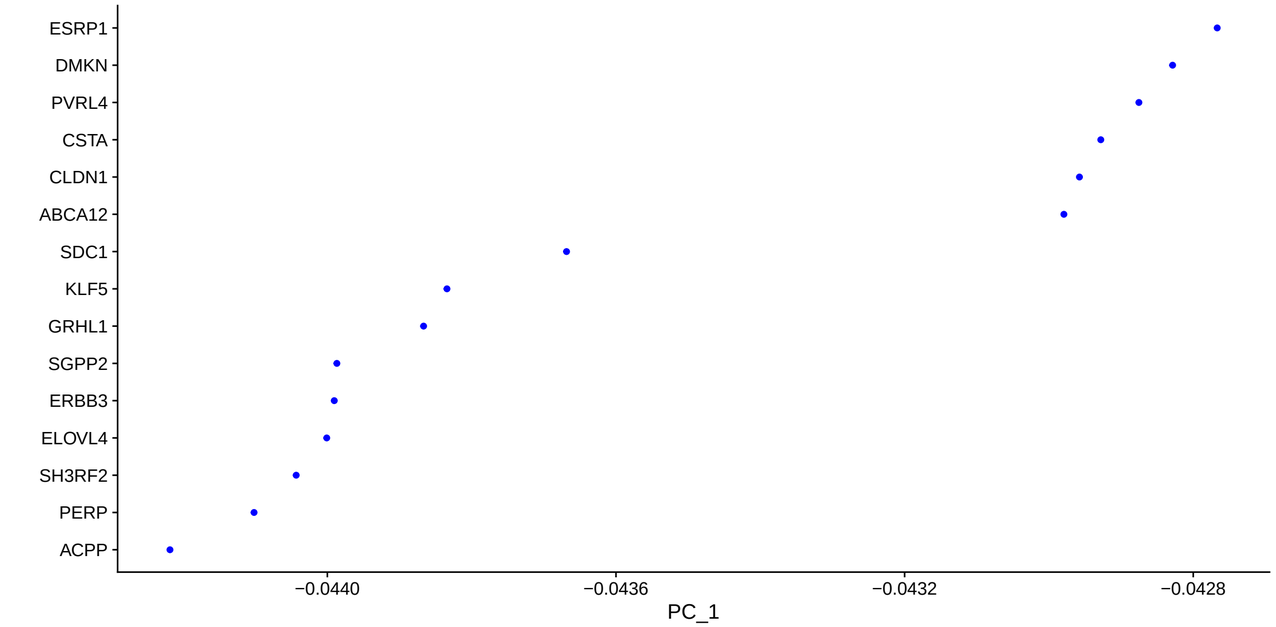

LC_BATCH
20
0
−20
PC2 (11.64%)
−25
0
25
PC1 (19.8%)
LCSET−12154
LCSET−12548
LCSET−12627
LCSET−17718
LCSET−18054
LCSET−18352
LCSET−20214
LCSET−20590
LCSET−23758
LCSET−26033

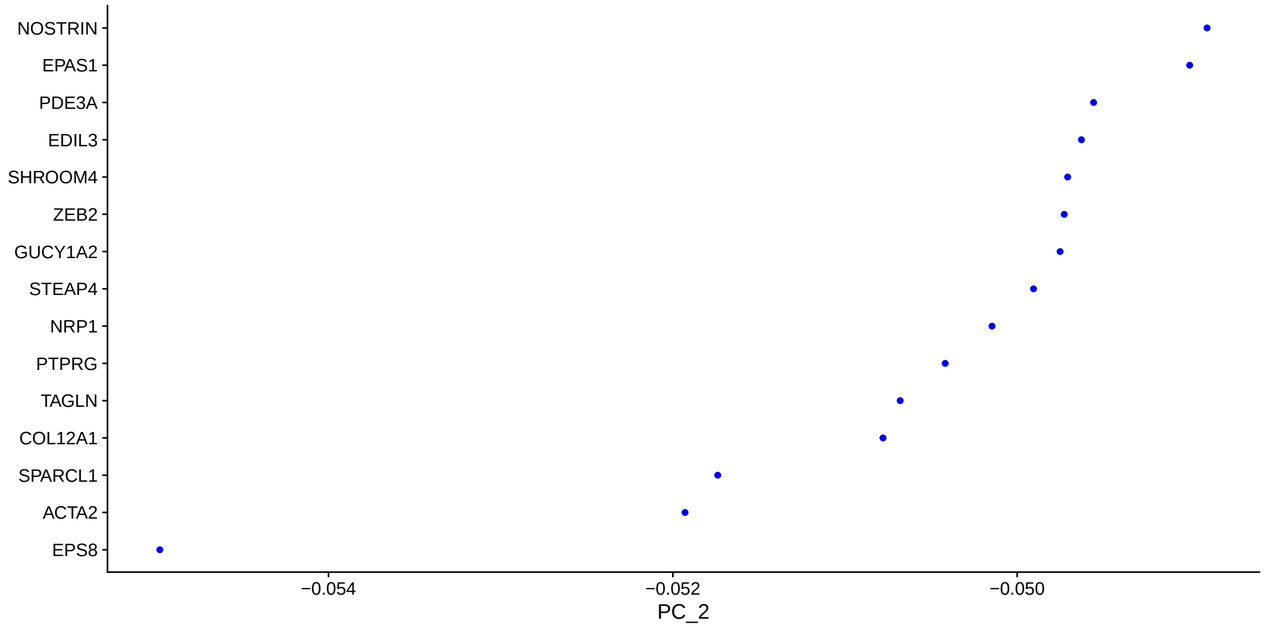
B

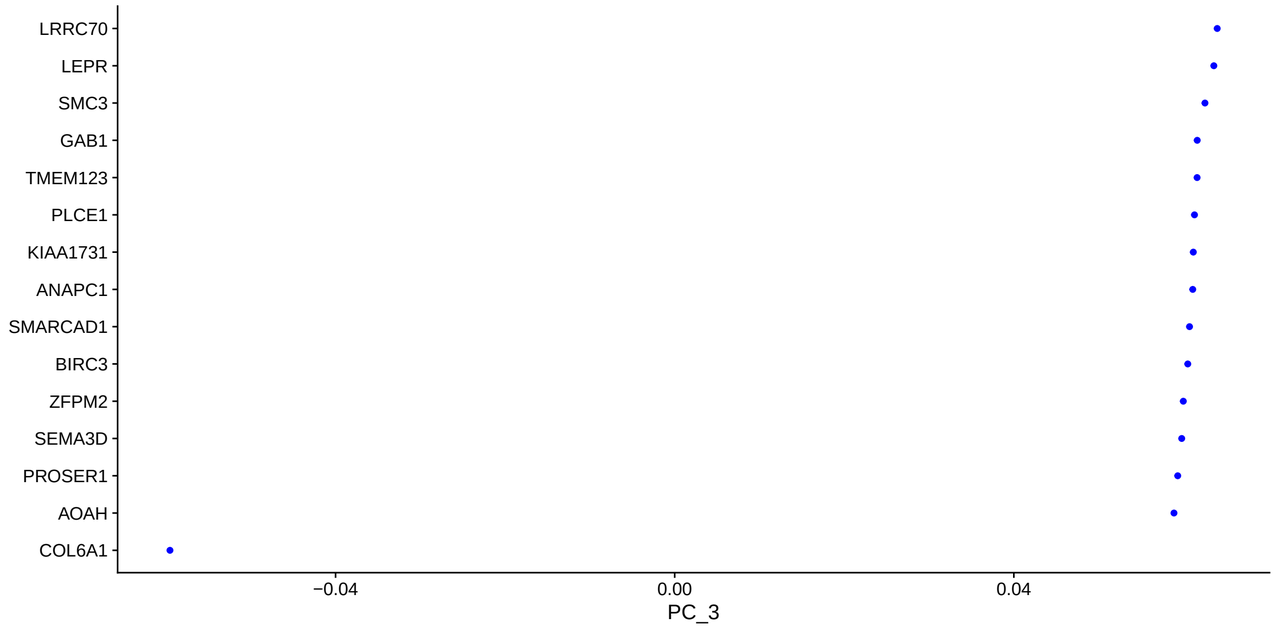

Primary Site (Recombined)
20
0
−20
PC2 (11.64%)
−25
0
25
PC1 (19.8%)
Breast (Cutaneous)
Breast (Parenchymal)
Extremities
Heart
Hepatobiliary
HNFS
Musculoskeletal
Other Visceral Organs
NA

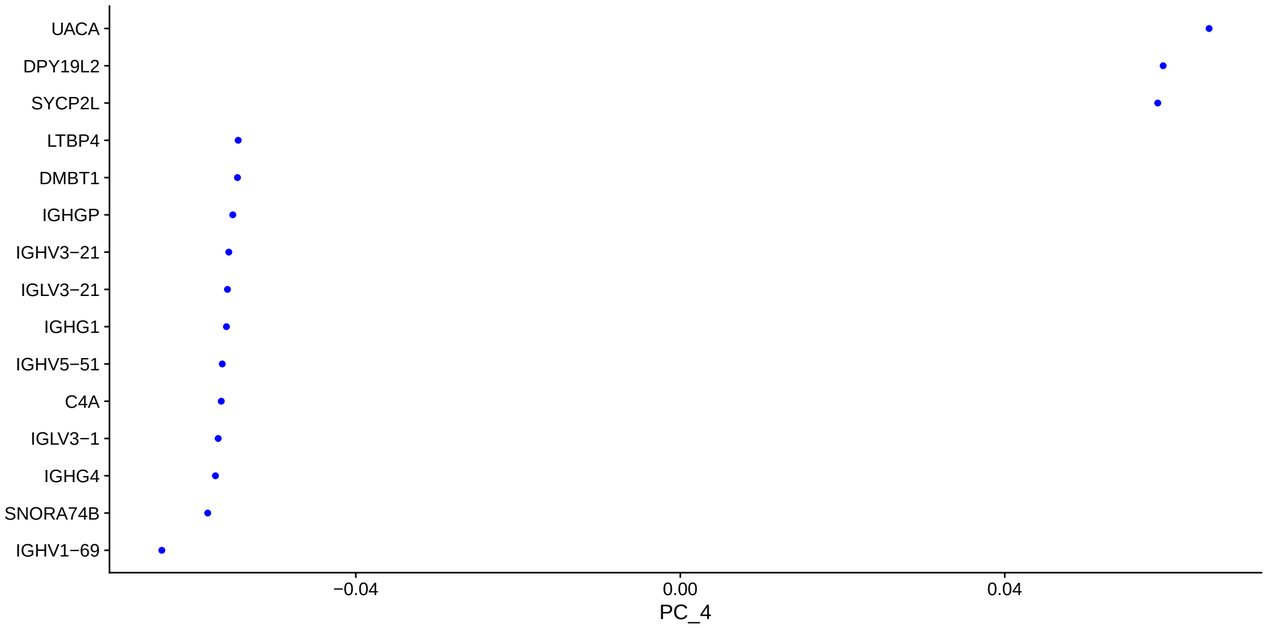

20
15
10
5
Standard Deviation
5
10
15
20
PC

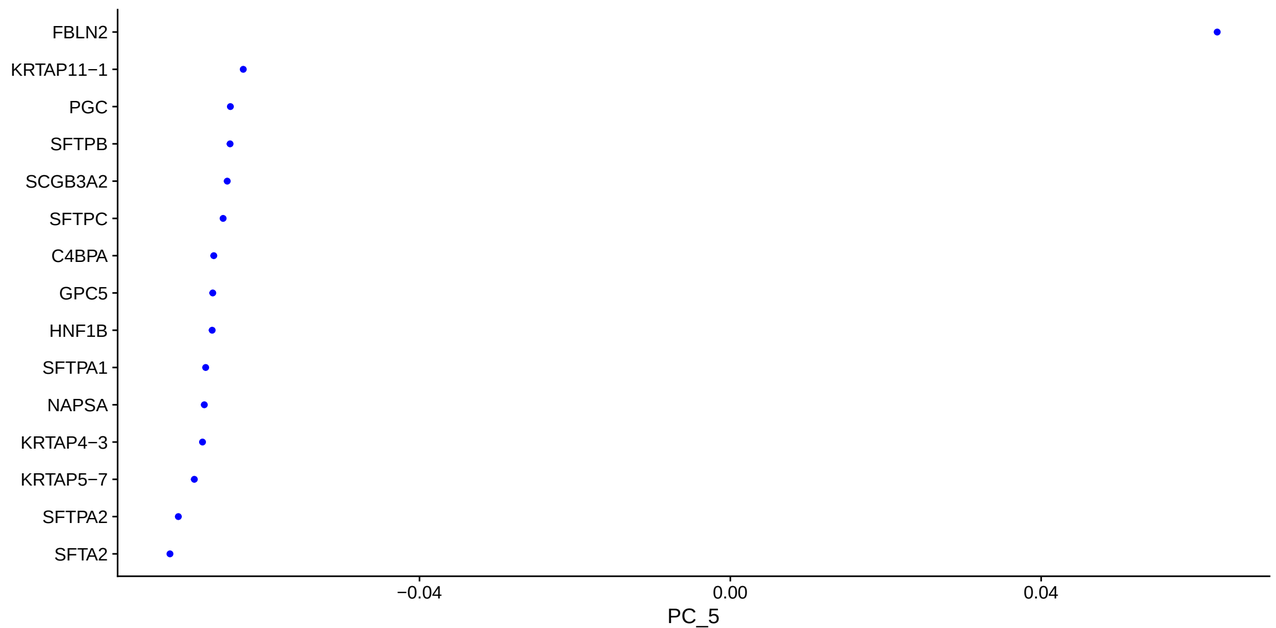

XIST
KRT1
KRT10
KRT5
FLG2
DSG1
FLG
DSP
KRT14
KRT2
Non−variable count: 19224
Variable count: 2000
Standardized Variance
20
10
0
5
10
30
Average Expression

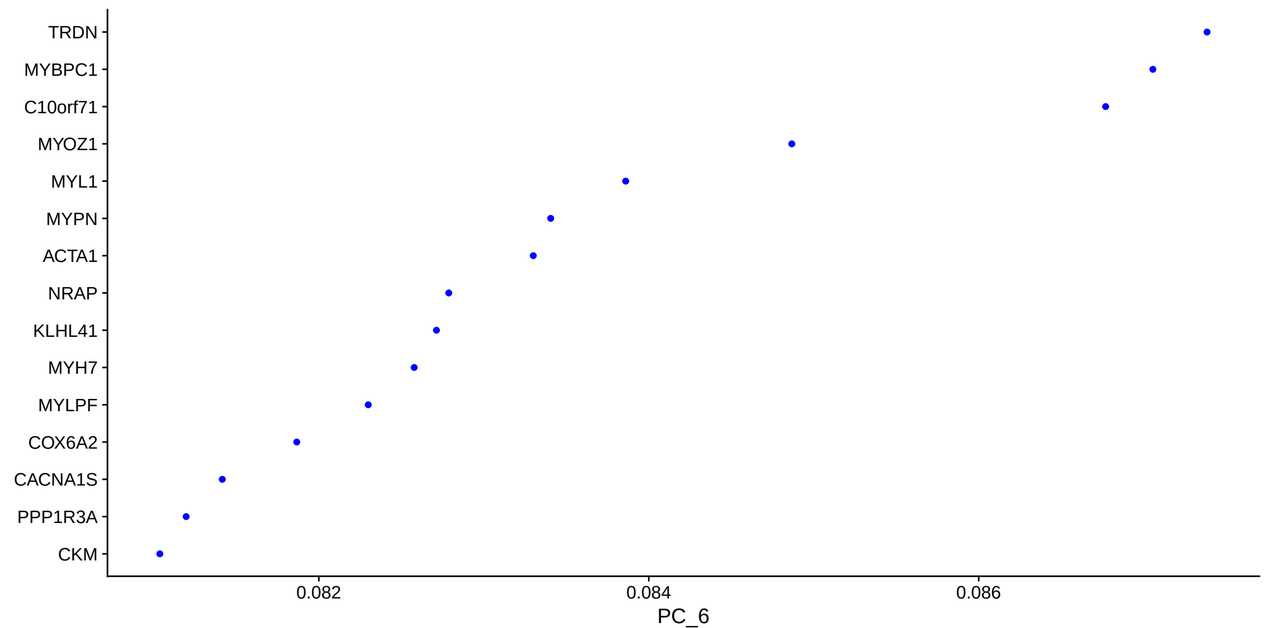

ESRP1
DMKN
PVRL4
CSTA
CLDN1
ABCA12
SDC1
KLF5
GRHL1
SGPP2
ERBB3
ELOVL4
SH3RF2
PERP
ACPP
−0.0440
−0.0436
−0.0432
−0.0428
PC_1

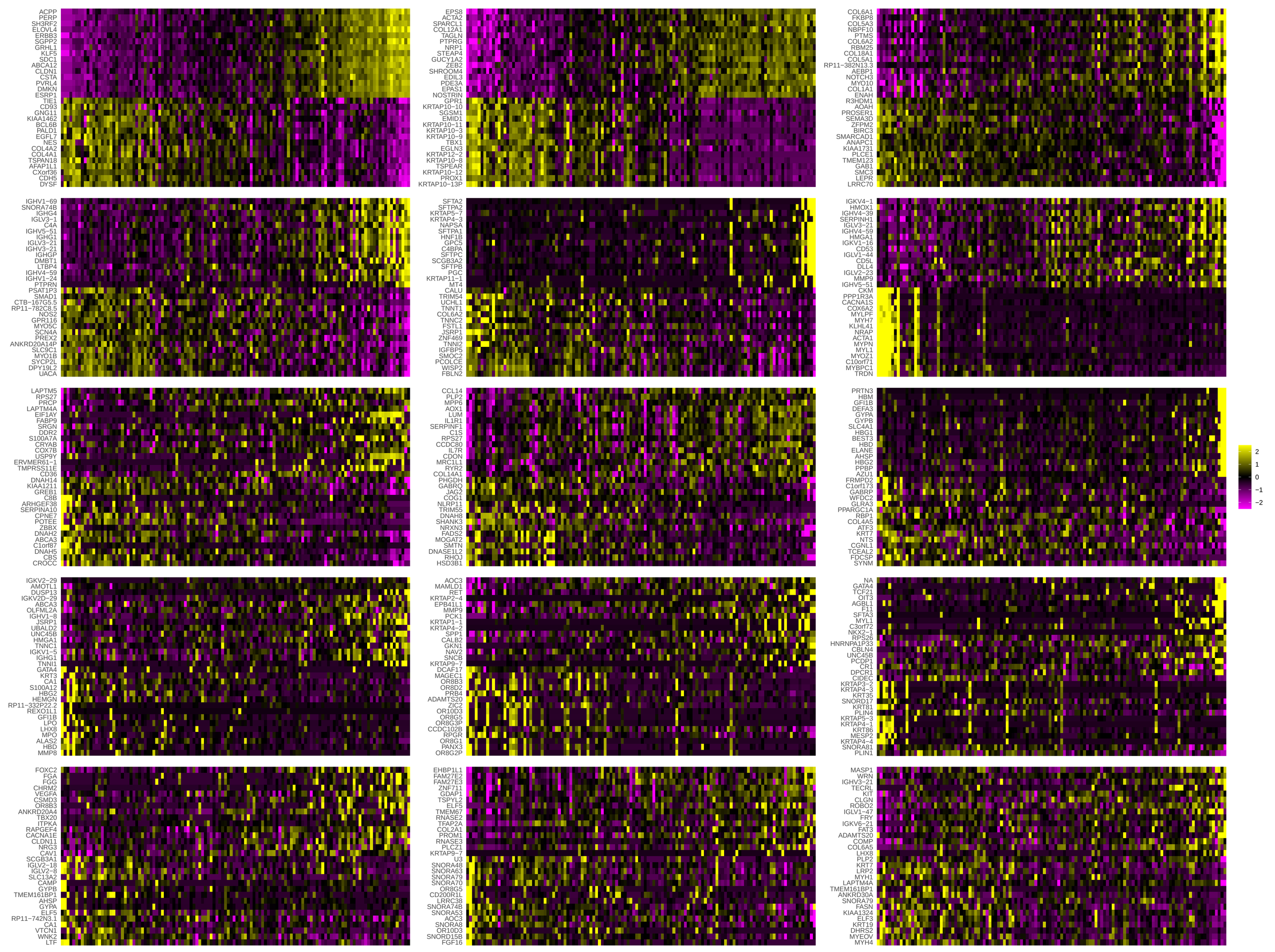

NOSTRIN
EPAS1
PDE3A
EDIL3
SHROOM4
ZEB2
GUCY1A2
STEAP4
NRP1
PTPRG
TAGLN
COL12A1
SPARCL1
ACTA2
EPS8
−0.054
−0.052
−0.050
PC_2


LRRC70
LEPR
SMC3
GAB1
TMEM123
PLCE1
KIAA1731
ANAPC1
SMARCAD1
BIRC3
ZFPM2
SEMA3D
PROSER1
AOAH
COL6A1
−0.04
0.00
0.04
PC_3


UACA
DPY19L2
SYCP2L
LTBP4
DMBT1
IGHGP
IGHV3−21
IGLV3−21
IGHG1
IGHV5−51
C4A
IGLV3−1
IGHG4
SNORA74B
IGHV1−69
−0.04
0.00
0.04
PC_4


FBLN2
KRTAP11−1
PGC
SFTPB
SCGB3A2
SFTPC
C4BPA
GPC5
HNF1B
SFTPA1
NAPSA
KRTAP4−3
KRTAP5−7
SFTPA2
SFTA2
−0.04
0.00
0.04
PC_5


TRDN
MYBPC1
C10orf71
MYOZ1
MYL1
MYPN
ACTA1
NRAP
KLHL41
MYH7
MYLPF
COX6A2
CACNA1S
PPP1R3A
CKM
0.082
0.084
0.086
PC_6


ACPP
PERP
SH3RF2
ELOVL4
ERBB3
SGPP2
GRHL1
KLF5
SDC1
ABCA12
CLDN1
CSTA
PVRL4
DMKN
ESRP1
TIE1
CD93
GNG11
KIAA1462
BCL6B
PALD1
EGFL7
NES
COL4A2
COL4A1
TSPAN18
AFAP1L1
CXorf36
CDH5
DYSF
EPS8
ACTA2
SPARCL1
COL12A1
TAGLN
PTPRG
NRP1
STEAP4
GUCY1A2
ZEB2
SHROOM4
EDIL3
PDE3A
EPAS1
NOSTRIN
GPR1
KRTAP10−10
SGSM1
EMID1
KRTAP10−11
KRTAP10−3
KRTAP10−9
TBX1
EGLN3
KRTAP12−2
KRTAP10−8
TSPEAR
KRTAP10−12
PROX1
KRTAP10−13P
COL6A1
FKBP8
COL5A3
NBPF10
PTMS
COL6A2
RBM25
COL18A1
COL5A1
RP11−382N13.3
AEBP1
NOTCH3
MYO10
COL1A1
ENAH
R3HDM1
AOAH
PROSER1
SEMA3D
ZFPM2
BIRC3
SMARCAD1
ANAPC1
KIAA1731
PLCE1
TMEM123
GAB1
SMC3
LEPR
LRRC70
IGHV1−69
SNORA74B
IGHG4
IGLV3−1
C4A
IGHV5−51
IGHG1
IGLV3−21
IGHV3−21
IGHGP
DMBT1
LTBP4
IGHV4−59
IGHV1−24
PTPRN
PSAT1P3
SMAD1
CTB−167G5.5
RP11−782C8.5
NOS2
GPR116
MYO5C
SCN4A
PREX2
ANKRD20A14P
SLC9C1
MYO1B
SYCP2L
DPY19L2
UACA
SFTA2
SFTPA
KRTAP5−7
KRTAP4−3
NAPSA
SFTPA1
HNF1B
GPC5
C4BPA
SFTPC
SCGB3A2
SFTPB
PGC
KRTAP11−1
MT4
CALU
TRIM54
UCHL1
TNNT1
COL6A2
TNNC2
FSTL1
JSRP1
ZNF469
TNNI2
IGFBP5
SMOC2
PCOLCE
WISP2
FBLN2
IGKV4−1
HMOX1
IGHV4−39
SERPINH1
IGLV3−21
IGHV4−59
HMGA1
IGKV1−16
CD53
IGLV1−44
CD5L
DLL4
IGLV2−23
MMP9
IGHV5−51
CKM
PPP1R3A
CACNA1S
COX6A2
MYLPF
MYH7
KLHL41
NRAP
ACTA1
MYPN
MYL1
MYOZ1
C10orf71
MYBPC1
TRDN
LAPTM5
RPS27
PRCP
LAPTM4A
EIF1AY
FABP9
SRGN
DDR2
S100A7A
CRYAB
COX7B
USP9Y
ERVMER61−1
TMPRSS11E
CD36
DNAH14
KIAA1211
GREB1
C8B
ARHGEF38
SERPINA10
CPNE7
POTEE
ZBBX
DNAH2
ABCA3
C1orf87
DNAH5
CBS
CROCC
CCL14
PLP2
MPP6
AOX1
LUM
IL1R1
SERPINF1
C1S
RPS27
CCDC80
IL7R
CDON
MRC1L1
RYR2
COL14A1
PHGDH
GABRQ
JAG2
COG1
NLRP11
TRIM55
DNAH8
SHANK3
NRXN3
FADS2
MOGAT2
SMTN
DNASE1L2
RHOJ
HSD3B1
PRTN3
HBM
GFI1B
DEFA3
GYPA
GYPB
SLC4A1
HBG1
BEST3
HBD
ELANE
AHSP
HBG2
PPBP
AZU1
FRMPD2
C1orf173
GABRP
WFDC2
GLRA3
PPARGC1A
RBP1
COL4A5
ATF3
KRT7
NTS
CGNL1
TCEAL2
FDCSP
SYNM
2
1
0
−1
−2
IGKV2−29
AMOTL1
DUSP13
IGKV2D−29
ABCA3
OLFML2A
IGHV1−8
JSRP1
UBALD2
UNC45B
HMGA1
TNNC1
IGKV1−5
IGHG1
TNNI1
GATA4
KRT3
CA1
S100A12
HBG2
HEMGN
RP11−332P22.2
REXO1L1
GFI1B
LPO
LHX8
MPO
ALAS2
HBD
MMP8
AOC3
MAMLD1
RET
KRTAP2−4
EPB41L1
MMP9
PCK1
KRTAP1−1
KRTAP4−2
SPP1
CALB2
GKN1
NAV2
SNCB
KRTAP9−7
DCAF17
MAGEC1
OR8B3
OR8D2
PRB4
ADAMTS20
ZIC2
OR10D3
OR8G5
OR8G3P
CCDC102B
RPGR
OR8G1
PANX3
OR8G2P
NA
GATA4
TCF21
OIT3
AGBL1
F11
SFTA3
MYL1
C3orf72
NKX2−1
RPS26
HNRNPA1P33
CBLN4
UNC45B
PCDP1
CR1
DPCR1
CIDEC
KRTAP3−2
KRTAP4−3
KRT35
SNORD17
KRT81
PLIN4
KRTAP5−3
KRTAP4−1
KRT86
MESP2
KRTAP4−4
SNORA81
PLIN1
FOXC2
FGA
FGG
CHRM2
VEGFA
CSMD3
OR8B3
ANKRD20A4
TBX20
ITPKA
RAPGEF4
CACNA1E
CLDN11
NRG3
CAV1
SCGB3A1
IGLV2−18
IGLV2−8
SLC13A2
CAMP
GYPB
TMEM161BP1
AHSP
GYPA
ELF5
RP11−742N3.1
CA1
VTCN1
WNK2
LTF
EHBP1L1
FAM27E2
FAM27E3
ZNF711
GDAP1
TSPYL2
ELF5
TMEM67
RNASE2
TFAP2A
COL2A1
PROM1
RNASE3
PLCZ1
KRTAP9−7
U3
SNORA48
SNORA63
SNORA79
SNORA70
OR8G5
CD200R1L
LRRC38
SNORA74B
SNORA53
AOC3
SNORA8
OR10D3
SNORD15B
FGF16
MASP1
WRN
IGHV3−21
TECRL
KIT
CLGN
ROBO2
IGLV1−47
FRY
IGKV6−21
FAT3
ADAMTS20
COMP
COL6A5
LHX8
PLP2
KRT7
LRP2
MYH1
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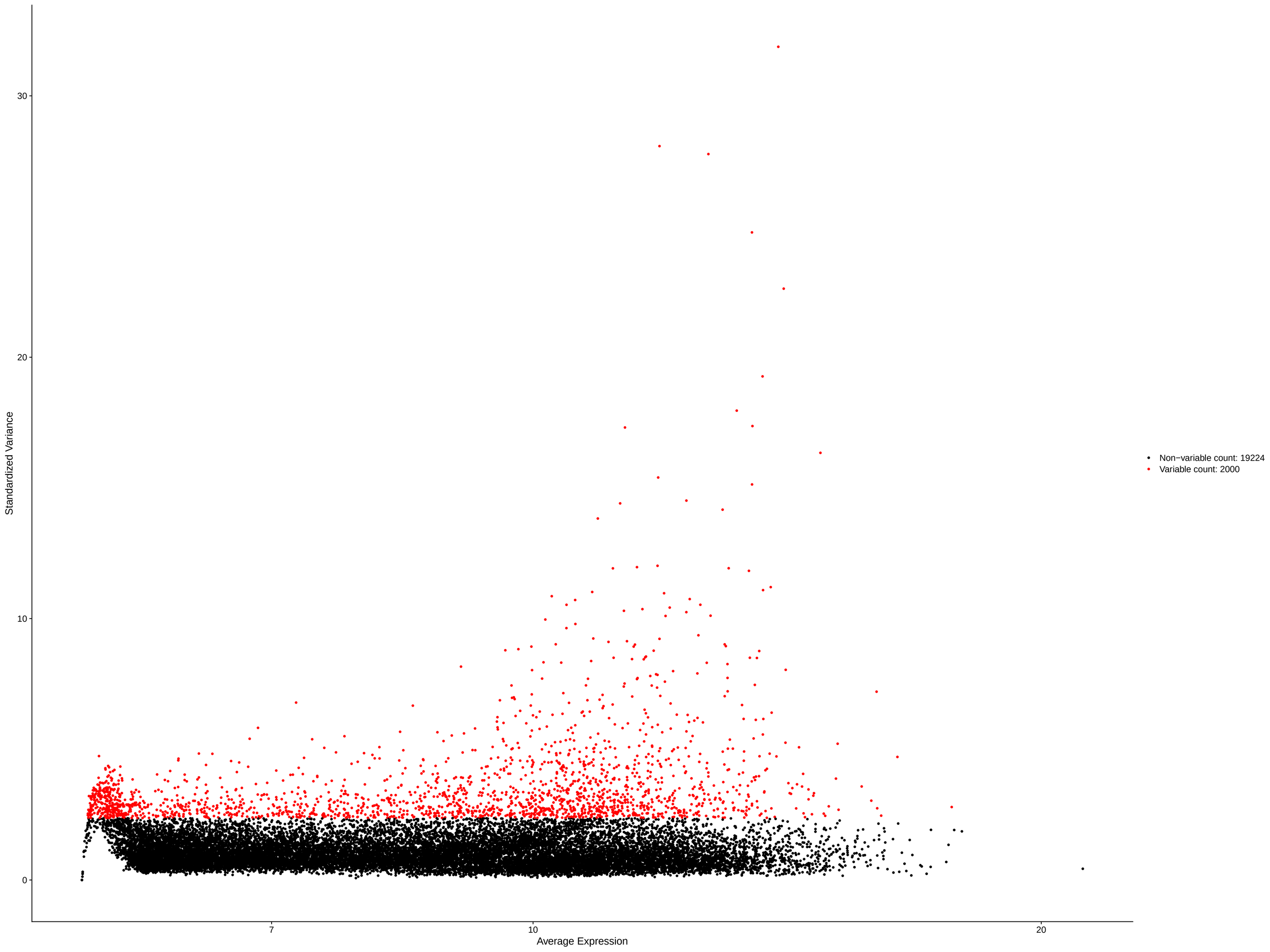
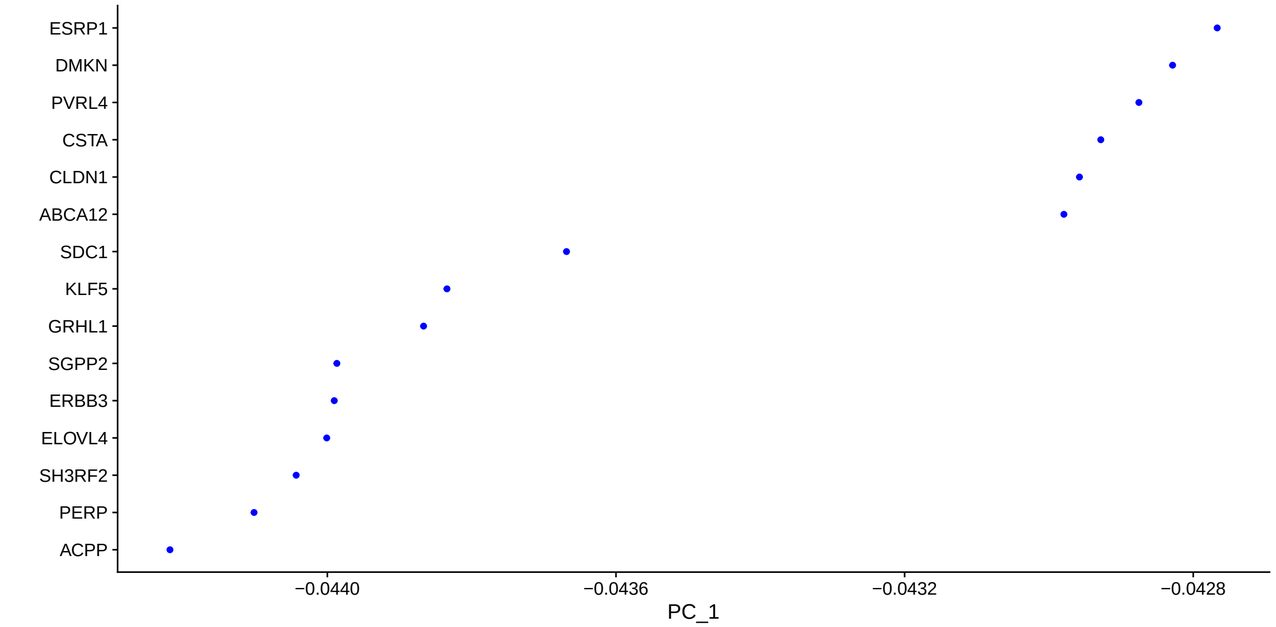
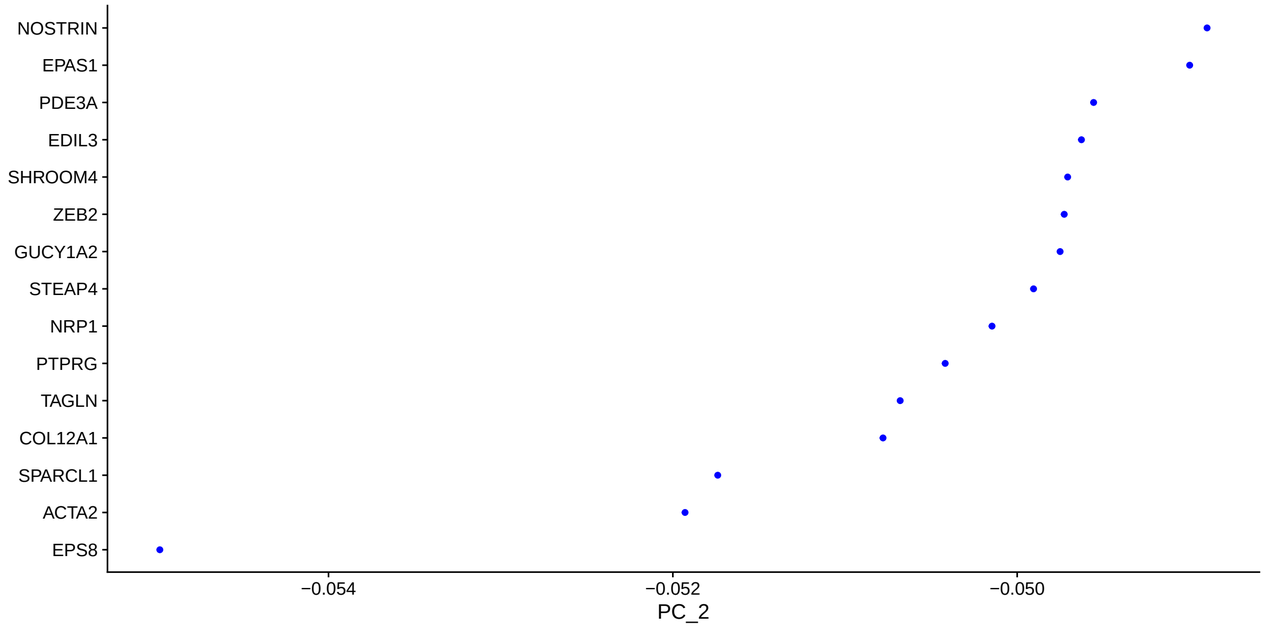
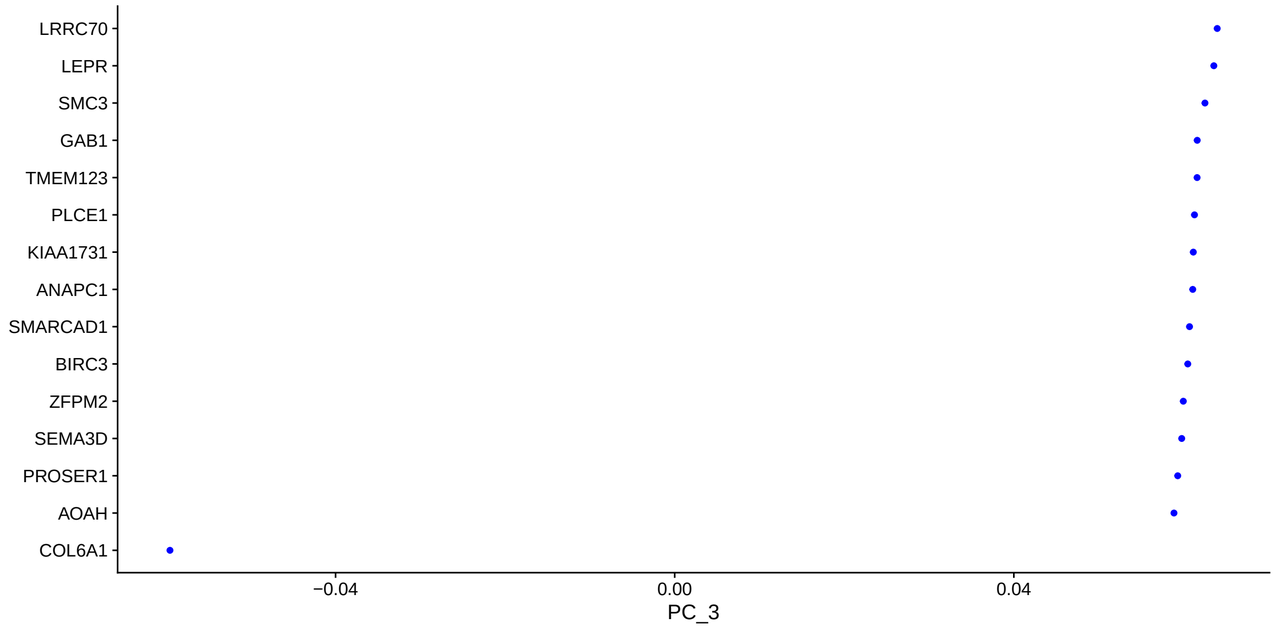
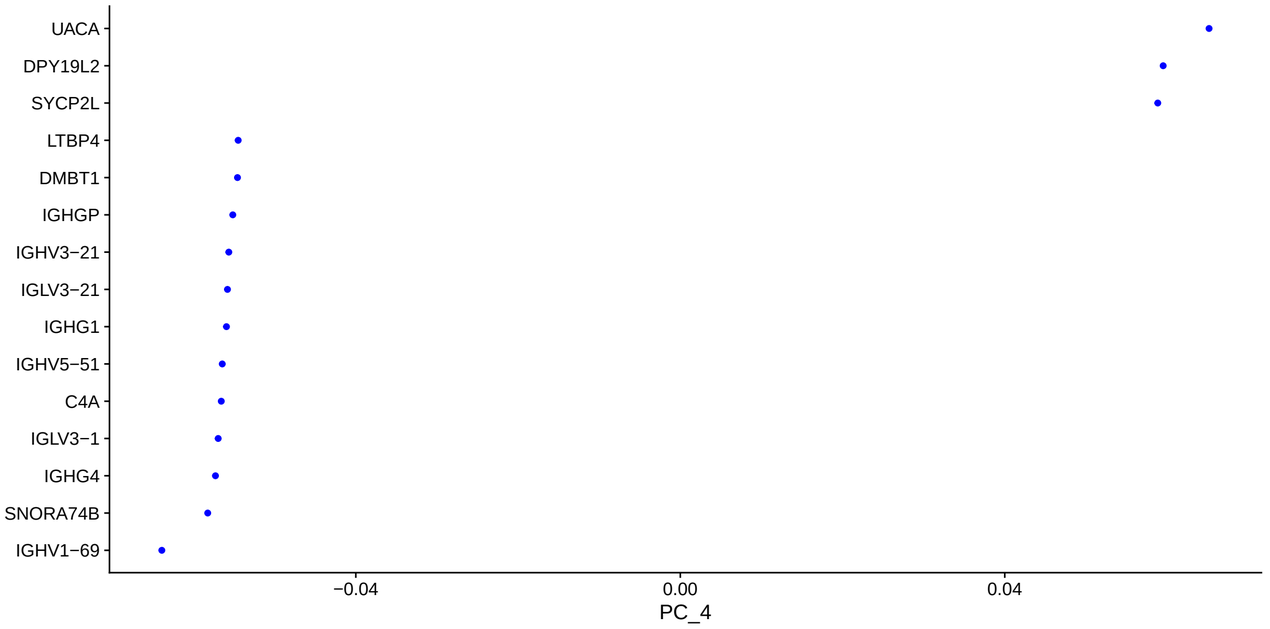
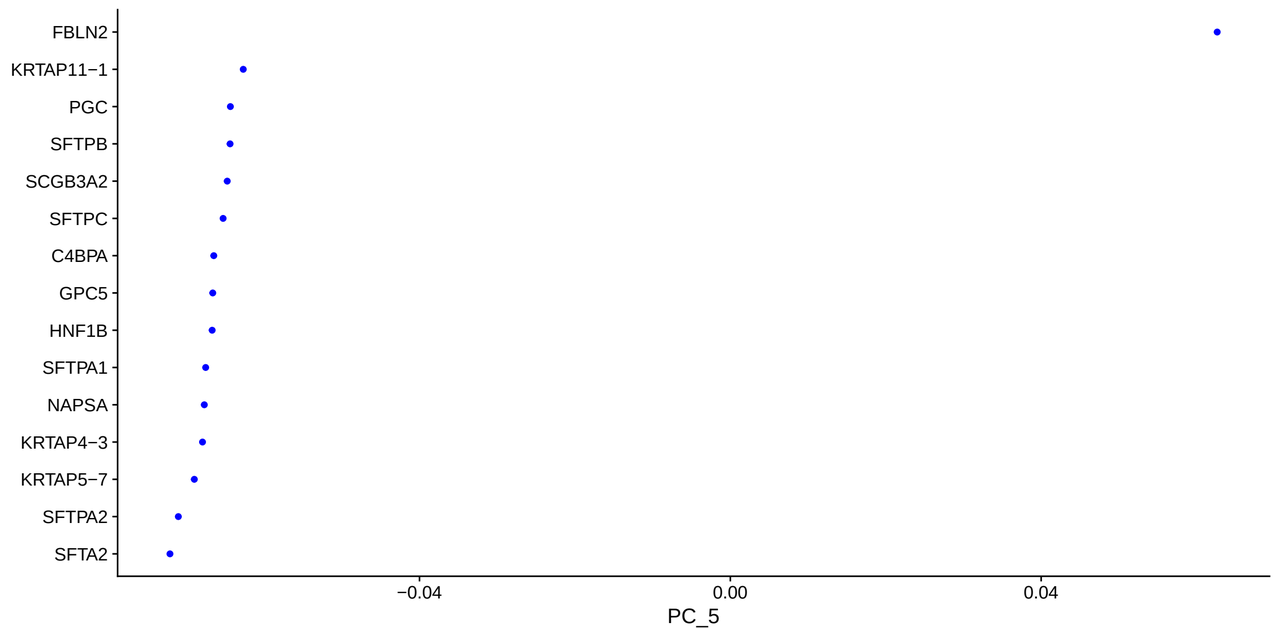

Standardized Variance
30
20
10
0
7
10
20
Average Expression
Non−variable count: 19224
Variable count: 2000

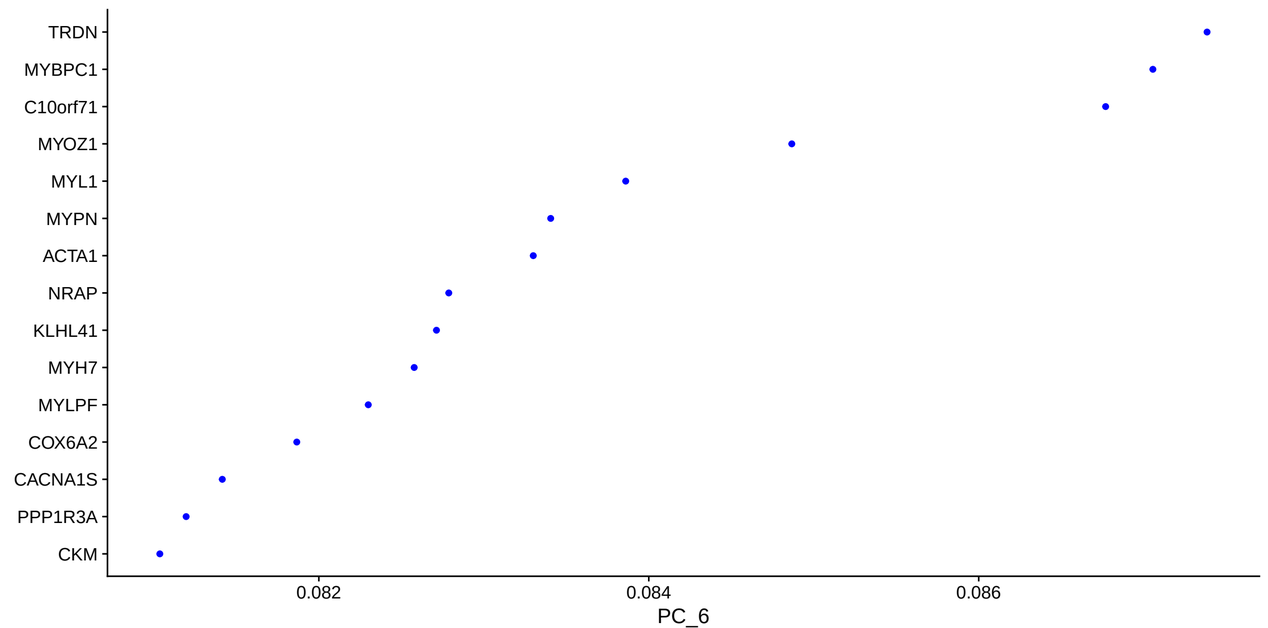

ESRP1
DMKN
PVRL4
CSTA
CLDN1
ABCA12
SDC1
KLF5
GRHL1
SGPP2
ERBB3
ELOVL4
SH3RF2
PERP
ACPP
−0.0440
−0.0436
−0.0432
−0.0428
PC_1

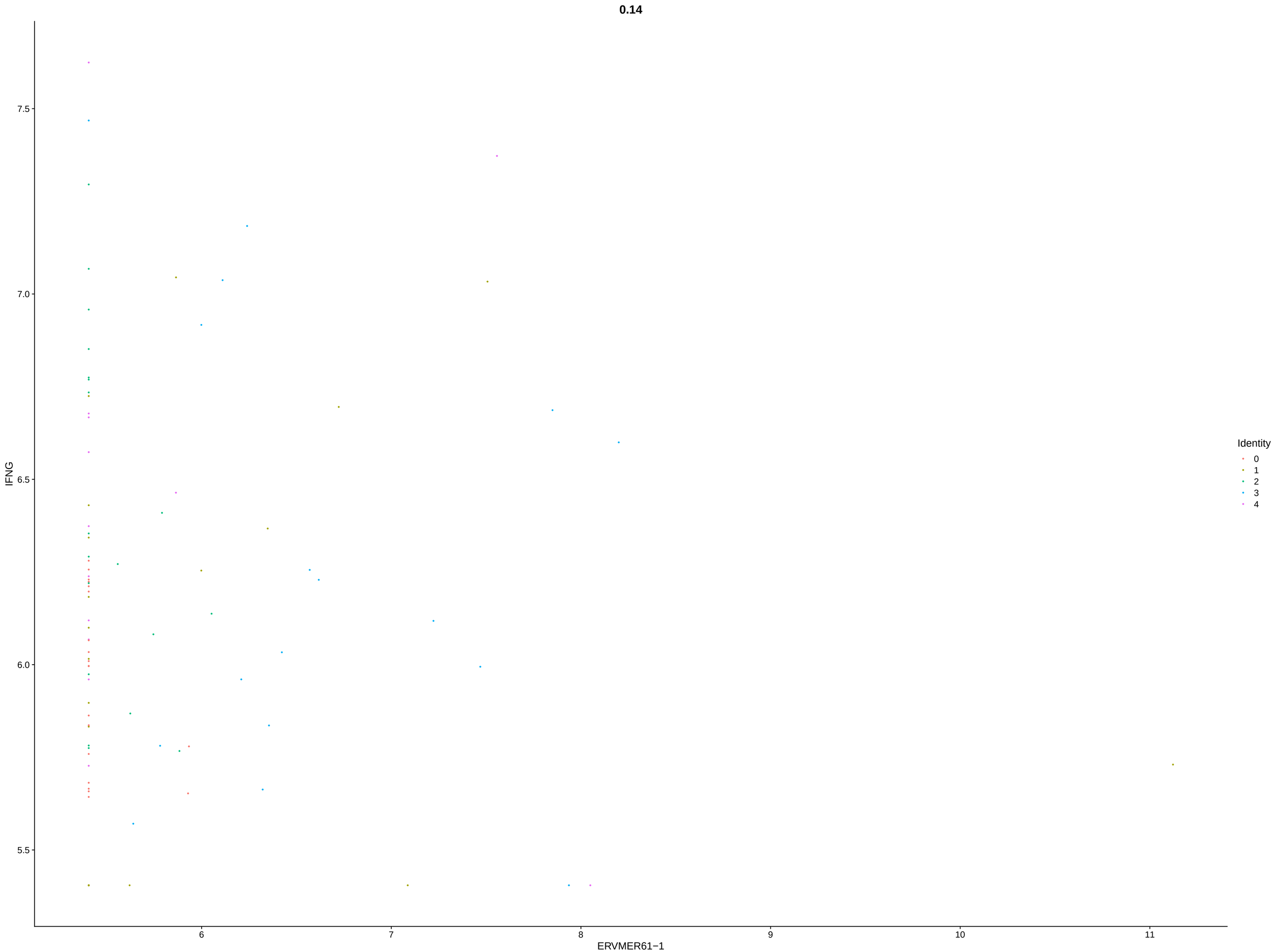

NOSTRIN
EPAS1
PDE3A
EDIL3
SHROOM4
ZEB2
GUCY1A2
STEAP4
NRP1
PTPRG
TAGLN
COL12A1
SPARCL1
ACTA2
EPS8
−0.054
−0.052
−0.050
PC_2

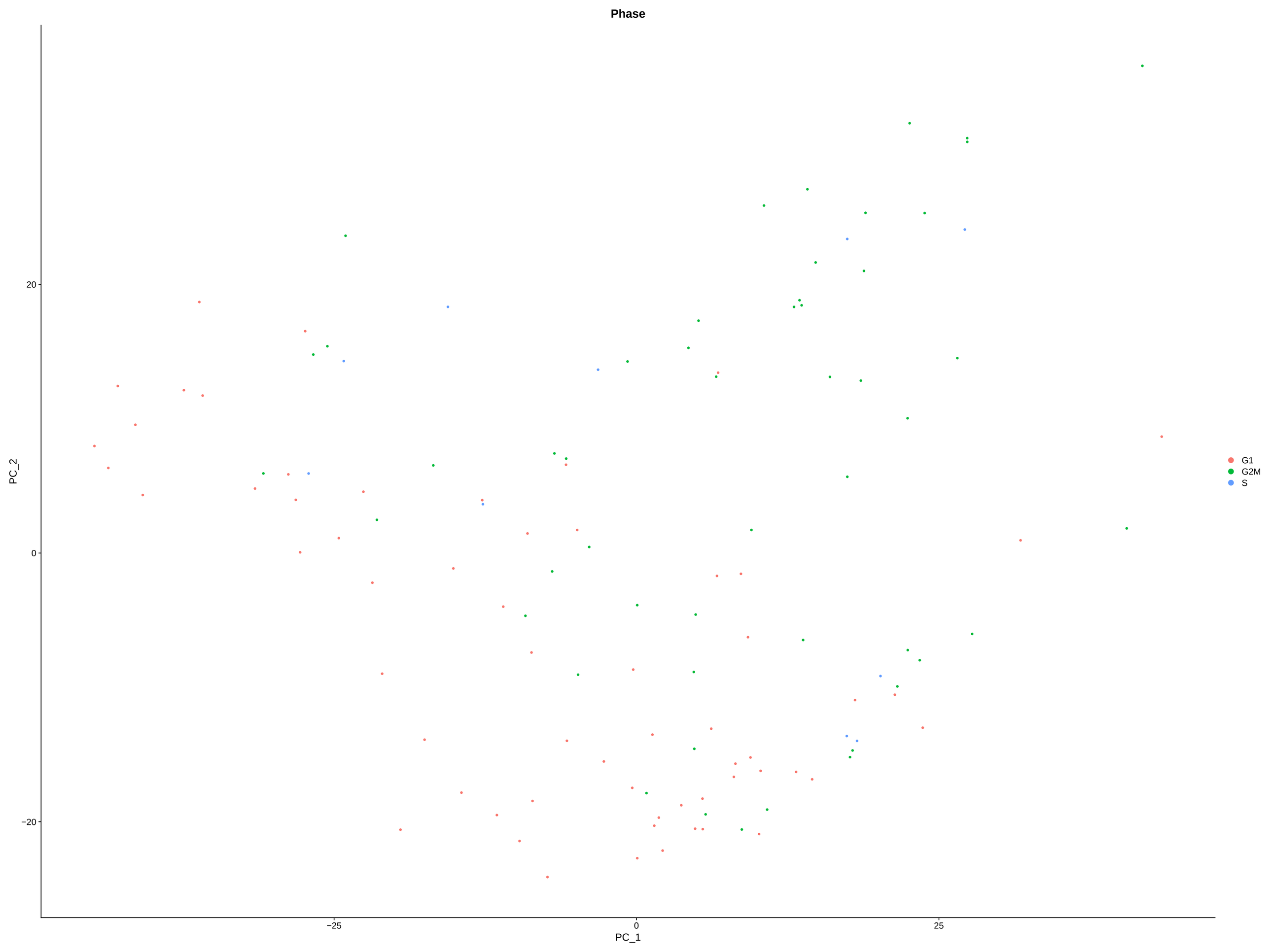

LRRC70
LEPR
SMC3
GAB1
TMEM123
PLCE1
KIAA1731
ANAPC1
SMARCAD1
BIRC3
ZFPM2
SEMA3D
PROSER1
AOAH
COL6A1
−0.04
0.00
0.04
PC_3


UACA
DPY19L2
SYCP2L
LTBP4
DMBT1
IGHGP
IGHV3−21
IGLV3−21
IGHG1
IGHV5−51
C4A
IGLV3−1
IGHG4
SNORA74B
IGHV1−69
−0.04
0.00
0.04
PC_4


FBLN2
KRTAP11−1
PGC
SFTPB
SCGB3A2
SFTPC
C4BPA
GPC5
HNF1B
SFTPA1
NAPSA
KRTAP4−3
KRTAP5−7
SFTPA2
SFTA2
−0.04
0.00
0.04
PC_5


TRDN
MYBPC1
C10orf71
MYOZ1
MYL1
MYPN
ACTA1
NRAP
KLHL41
MYH7
MYLPF
COX6A2
CACNA1S
PPP1R3A
CKM
0.082
0.084
0.086
PC_6


0.14
IFNG
7.5
7.0
6.5
6.0
5.5
6
7
8
9
10
11
ERVMER61−1
Identity
0
1
2
3
4


Phase
G1
G2M
S
PC_2
20
0
−20
−25
0
25
PC_1